# Review Midterm 2 Math 2270

Ex 1 Compute eigenvalues and eigenvectors for $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$, $\begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$, $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$.

Ex 2 For $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ as above, find $S$, $S^{-1}$ and compute $S^{-1}AS$.

Ex 3 Give the defn of eigenvalue / eigenvector.

Ex 4 Give an example of a 2×2 matrix with 0, 1, 2 eigenvalues.

Ex 5 Show that if $x$ is an eigenvector of eigenvalue $\lambda$ for $A$, then it is an eigenvector of eigenvalue $\lambda^n$ for $A^n$ $(n \in \mathbb{Z})$.

Ex 6 State the 3 fundamental properties of the determinant. Prove they hold for 2×2 matrices.

Ex 7 Find the area of the triangle with vertices $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$, & the volume of the parallelopiped spanned by these vectors.

Ex 8 Compute $\det(\lambda A)$, $\det(A^2)$, $\det(A^{-1})$, $\det(A^T)$

Ex 9 Show that $\det(A+B) \neq \det(A) + \det(B)$.

Ex 10 Explain why $\det(A)$ if two rows are equal and the $\det(A)$ is unchanged if you add a multiple of one row to the ~~another~~ other row.

Ex 11 State the cofactor formula for $A^{-1}$ and compute the 2,3 entry of $\begin{pmatrix} 1 & t & 0 \\ t & 1 & t \\ 0 & t & 1 \end{pmatrix}^{-1}$

Ex 12 Use Cramer's rule to find $z$ $\begin{cases} 2x - 3y + z = 1 \\ x + y - 2z = 1 \\ 3x - 2y + 2z = 1 \end{cases}$